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MySQL 8.0 Command Line CL  x  +  v
mysql> CREATE DATABASE CollegeDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE department (
  ->   Dept_ID INT PRIMARY KEY,
  ->   Dept_Name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.18 sec)

mysql> INSERT INTO department VALUES
  -> (1, 'CSE'),
  -> (2, 'ECE'),
  -> (3, 'EEE'),
  -> (4, 'MECH');
Query OK, 4 rows affected (0.06 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE faculty (
  ->   Faculty_ID INT PRIMARY KEY,
  ->   Faculty_Name VARCHAR(30),
  ->   Dept_ID INT
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO faculty VALUES
  -> (201, 'Ramesh', 1),
  -> (202, 'Meena', 2),
  -> (203, 'Suresh', 3),
  -> (204, 'Lakshmi', 4);
Query OK, 4 rows affected (0.06 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT faculty.Faculty_ID,
  ->   faculty.Faculty_Name,
  ->   department.Dept_ID,
  ->   department.Dept_Name
  -> FROM faculty
  -> CROSS JOIN department
  -> WHERE faculty.Dept_ID = department.Dept_ID;
```

Faculty_ID	Faculty_Name	Dept_ID	Dept_Name
201	Ramesh	1	CSE
202	Meena	2	ECE
203	Suresh	3	EEE
204	Lakshmi	4	MECH

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4 rows in set (0.00 sec)

mysql> CREATE TABLE student (
  ->     Student_ID INT PRIMARY KEY,
  ->     Student_Name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO student VALUES
  -> (301, 'Arun'),
  -> (302, 'Bhavya'),
  -> (303, 'Charan'),
  -> (304, 'Deepa');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE score (
  ->     Student_ID INT,
  ->     Marks INT,
  ->     Course_No INT
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO score VALUES
  -> (301, 85, 101),
  -> (302, 78, 102),
  -> (303, 92, 103);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT student.Student_Name,
  ->         score.Marks,
  ->         score.Course_No
  -> FROM student
  -> LEFT JOIN score
  -> ON student.Student_ID = score.Student_ID
  -> WHERE score.Marks IS NULL;
+-----+-----+-----+
| Student_Name | Marks | Course_No |
+-----+-----+-----+
| Deepa       | NULL  | NULL      |
+-----+-----+-----+
1 row in set (0.01 sec)

mysql> |

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1 row in set (0.01 sec)

mysql> CREATE TABLE course (
->     Course_ID INT PRIMARY KEY,
->     Course_Name VARCHAR(50),
->     Faculty_ID INT
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO course VALUES
-> (401, 'Data Structures', 201),
-> (402, 'Database Systems', 201),
-> (403, 'Digital Communication', 202),
-> (404, 'Power Systems', 203);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT course.Course_Name
-> FROM course
-> JOIN faculty
-> ON course.Faculty_ID = faculty.Faculty_ID
-> JOIN department
-> ON faculty.Dept_ID = department.Dept_ID
-> WHERE department.Dept_Name = 'CSE'
-> AND faculty.Faculty_Name = 'Ramesh';
+-----+
| Course_Name |
+-----+
| Data Structures |
| Database Systems |
+-----+
2 rows in set (0.00 sec)

mysql> |
```

